Assignment - 1

Multi agent Architecture design for Housing Trends Application

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1 Problem and Design Objective

The system analyzes residential listings matching user-specified criteria and reports price movement over a twelve-month period. A single LLM call cannot safely handle this task: retrieval sufficiency, statistical computation, and anomaly diagnosis are distinct competencies with different failure modes. Critically, some failures thin samples, outlier-driven spikes are only detectable after a first computation is attempted.

We employ a multi-agent pipeline with explicit feedback loops and conditional routing. The architecture must exhibit: (a) a genuine downstream-to-upstream feedback loop where one agent's judgment changes what an upstream agent does next; (b) a conditional/iterative stop-or-continue control step based on a measurable signal; and (c) a dependency where one agent's output determines not merely the data another agent receives, but how that agent processes it.

2 Agent Architecture

We decompose the analysis into six specialized agents, each with a single responsibility and clear failure mode. No agent both diagnoses and corrects its own output.

2.1 Input Handler (Dual Mode)

Accepts user input in two forms:

- **Structured input** (form submission): If the user wants to supply structured form input (price range, bed/bath, property type, ZIP, date range), a deterministic parser function runs. It provides form based GUI for users to enter input. Here it uses no agent.
- **Free-text input**: If the user instead wishes to give a free-text query ("affordable family homes near good schools in North Dallas") in a text box, a **Parser Agent** is invoked to resolve ambiguous intent into the same structured schema (ZIP list, price bounds, property filters, date range).

Both paths converge on one structured-criteria object. Downstream agents remain agnostic to input mode.

2.2 Scanner Agent

Retrieves listings matching the user defined criteria from a frozen, preprocessed dataset. For each ZIP-month bucket, it reports listings of properties.

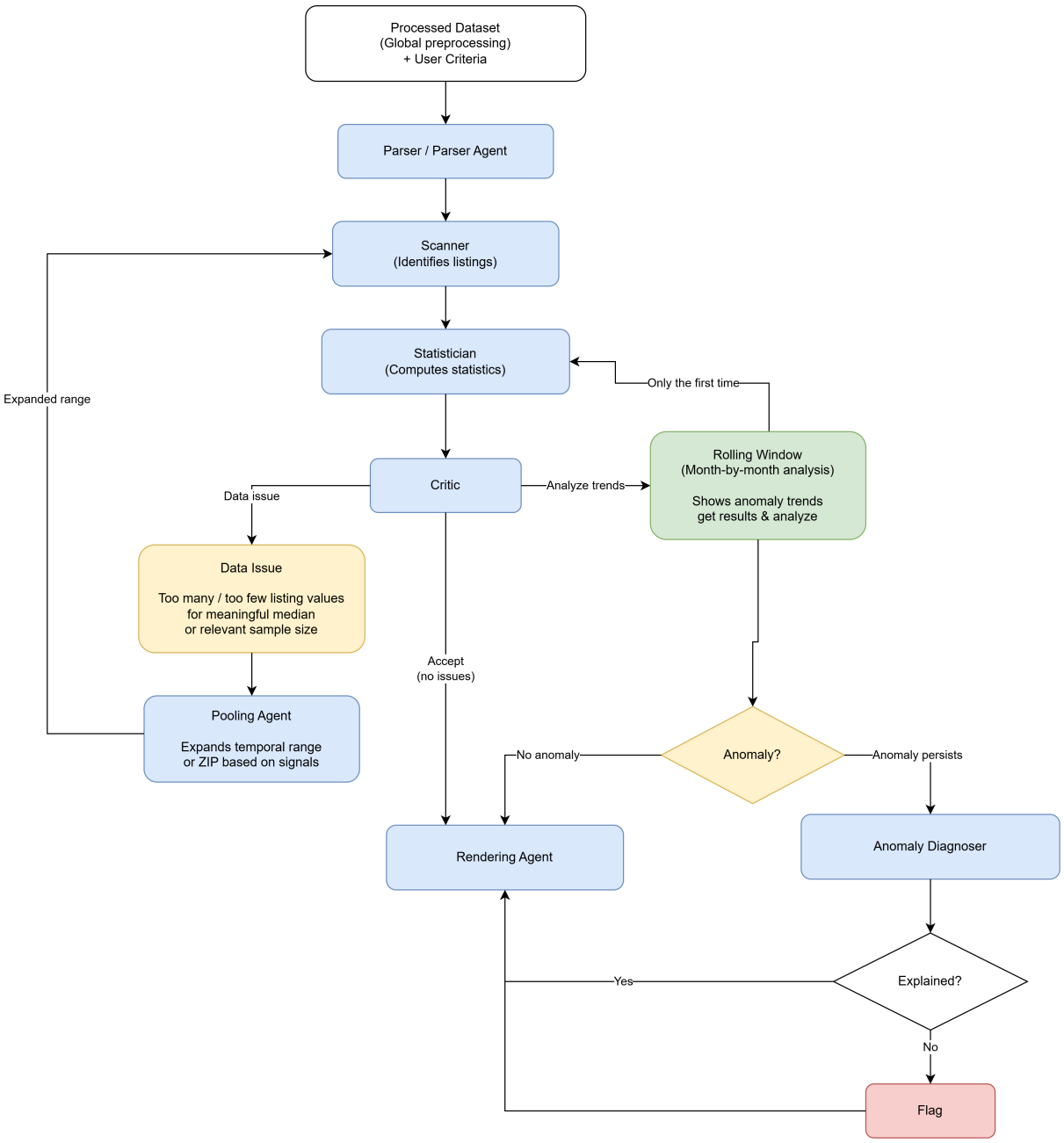


Figure 1: The multi-agent architecture diagram of the application

2.3 Statistician Agent

It is purely computational. It computes the requested statistic based on user criteria.(eg: median list price, price per sqft by month and ZIP) over whatever listing set and method it is told to use. It never inspects or judges its own output. It only recomputes when explicitly told to do so by **Critic Agent** and given a specific method change.

It also calculates month by month percentage change over 12 months. This is used by Critic Agent downstream to detect anomalous trends in data variations.

2.4 Critic Agent

The primary reviewer. Routes to one of three outcomes: **Data Issue**, **Analysis Issue**, or **Accept**.

1. **Data Issue**: By doing coverage check, it analyses if the supporting sample is adequate. We set a minimum threshold (say, ≥ 20 valid listings) per ZIP-month bucket. It invokes **Pooling Agent**.

2. **Analysis Issue**: Identifies anomalous trends by rolling month-over-month comparison, flagging big jumps in median price or in active-listing count. It then calls and sends data back to the **Statistician Agent** for recomputation. It only happens once. If the anomalous trend still persists, it is marked as an anomaly and is sent to the **Anomaly Diagnoser Agent**.

3. **Accept**: If no data issues or anomalous trends found, it is sent to **Rendering Agent**.

2.5 Pooling Agent

Invoked only when Critic Agent identifies a **Data Issue** problem to compute results. It does **Scope Expansion** i.e., Widen ZIP radius (add adjacent ZIPs), extend the date window, or relax non-protected filters (e.g., property type).

Outputs a revised query sent back to Scanner, or a verdict on how the scope should be expanded to increase listings by relaxing user-defined criteria. It stops once it meets the minimum threshold or when it reaches a certain number of retries.

2.6 Anomaly Diagnoser Agent

Investigates trends flagged as anomalous after Statistician's first recompute. Examines:

- Changes in property type, bedroom/bathroom counts, and square footage distribution.
- ZIP-level composition shifts (e.g., switch from Central to South Dallas).
- Sample size changes and extreme-value sensitivity.

It then tries to explain the anomalous trend by adding explanatory text. This is then sent to the **Rendering Agent**.

2.7 Rendering Agent

Displays the final trend, coverage metrics, warnings, effective filters, and agent decision log. Cannot alter statistics. It is purely presentational.

3 Control Flow and Feedback Loops

3.1 Three-Way Routing from Critic: Feedback Cycles

3.1.1 Cycle A: Critic → Pooling → Scanner

Data Issue: Sample is thin or unevenly distributed. Send coverage profile to Pooling. Pooling revises the query, scanner re-executes and statistician recomputes on the new dataset.

3.1.2 Cycle B: Critic → Statistician (Analysis Retry)

Analysis Issue: If a month-over-month jump is flagged suspicious, it send the same record set to Statistician with instruction to recompute. If anomalous trend still persists, escalate to Anomaly Diagnoser.

4 Conditional / Iterative Steps

- **Coverage Loop (Cycle A: Critic → Pooling → Scanner)**: Maximum 3 rescans, only while number of property listings is less than the required threshold. If threshold is reached, stop immediately.
- **Anomaly Investigation (Conditional Anomaly Check)**:After an analysis retry, the Critic checks whether the suspicious trend still persists. If the anomaly is resolved after recomputation, the verified result is sent directly to the Rendering Agent. If the anomaly persists, the result is forwarded to the Anomaly Diagnoser for further investigation.

5 Genuine Dependency

Cycle A: Critic → Pooling → Scanner: The Critic's diagnosis determines the strategy used by the Pooling Agent:

- If the sample is small, Pooling expands the date range.
- If listings are concentrated in one ZIP code, Pooling adds nearby ZIP codes.
- If the property-type filter is too restrictive, Pooling may relax that non-protected filter.
- If the Scanner reports exhaustive coverage, Pooling stops further expansion.

The Pooling Agent then sends a revised query to the Scanner. Therefore, the Critic's output changes the search scope and retrieval strategy used by the Scanner.

6 Data Source

We use the **Kaggle USA Real Estate Dataset (2.2M listings)**, filtered to Dallas–Fort Worth metro area (ZIP codes 75xxx). The dataset is frozen locally as a CSV snapshot, containing per-listing records: price, bedrooms, bathrooms, square footage, ZIP, property type, and listing date. Preprocessing (non-agentic) removes duplicate records and listings with zero or negative prices. **Zillow Research ZHVI** is used to provide temporal values at a monthy granularity (these values are not available in Kaggle and are inferred from Zillow).